AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

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Instructions:

- Total Marks: **50 marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks, 1 for each question]

1- The keyword ..... is used to create an object in memory.

a- class
b- new
c- object
d- create

2- What is the value stored in the variable result after executing the following code segment?

```
string result = "Programming".ToLower().Insert(3, "X");
```

a- proXgramming
b- ProXgramming
c- progXramming
d- programmingX

3- The ..... property returns the number of elements in an array.

a- Count
b- Size
c- Length
d- Index

4- What is the output of the following code?

```
int[] nums = { 4, 6, 8, 10 };
Console.WriteLine(nums[2]);
```

a- 4
b- 6
c- 8
d- 10

5- The ..... class member can be accessed by code outside the class.

a- private
b- public
c- protected
d- hidden

6- What is the value of z after executing the following code segment?

```
int x = 40;
float y = 12.6f;
int z = x + Convert.ToInt32(y);
```

a- 52
b- 52.6
c- 53
d- 54

7- What is the output of the following code?

```
int[,] values = { { 3, 5 }, { 7, 9 } };
Console.WriteLine(values[0,1] * values[1,0]);
```

a- 15
b- 21
c- 35
d- 63

8- The method ..... checks whether a character is a letter.

a- char.IsLetter(ch)
b- char.CheckLetter(ch)
c- char.Letter(ch)
d- char.IsAlphabet(ch)

9- The class from which another class inherits is called .....

a- derived class
b- child class
c- base class
d- object class

10- When a property is used to assign a value to a private field, the ..... accessor is used.

a. get
b. set
c. value
d. return

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Part 2: Short Questions [18 marks]

Q 2: Give a comparison (with examples) between Abstract class and interface in C#. (6 Marks)

Q3: Trace the following C# code segment and show its output. **(6 Marks)**

```
string message = "M109 Final Exam - KSA Students";
Console.WriteLine(message.Contains("final"));
Console.WriteLine(message.IndexOf("KSA"));
Console.WriteLine(message.Remove(5, 11));
Console.WriteLine(message.ToLower());
var sb = new StringBuilder(message);
Console.WriteLine(sb.ToString());
sb.Replace("Exam", "25-26");
sb.Append(" Spring");
Console.WriteLine(sb.ToString());
```

Q4: A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself. The most efficient way to check if a number is prime is to test for divisibility by numbers from 2 up to the square root of the number.
Implement a C# function that accepts an integer and return if it is a prime number or not. (6 Marks)

Part 3: Problem Solving Questions [22 marks]

Q 5: Write a C# program using Windows Forms that processes products prices as follows: (12 Marks)

- Read product data (Product Name, Price) from a file named Products.txt.
- Store product names in one array and prices in another array.
- Calculate the average price of all products.
- Find the product with the highest price.

Consider the following form requirements:

- TextBox named txtProductsFile to hold the file name.
- Button named btnReadProducts to read the data from the file.
- Button named btnProductsReport to display:
- Number of products.
- Average price.
- Maximum price.
- Product name with the maximum price.
- Button named btnExit to close the application.

Write only the code of btnProductsReport and the necessary field declarations.

Q 6: (10 Marks)

A. Write the Employee class which includes: (4 Marks)

1. Two fields: EmployeeName and Salary.
2. A parameterized constructor.
3. Getters and setters.
4. Override ToString() method.

B. Write the Manager class as a child class of Employee which includes: (3 Marks)

1. One additional field: Department

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2. Parameterized constructor.
3. Override ToString() method.

C. Write the Main method to: (3 Marks)
1. Create one Employee object.
2. Create one Manager object.
3. Store them in a List.
4. Display their information using ToString().

End of the questions.



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